# An Experimental Macroscopic Study of Secret Religions During the Jiaqing Period of the Qing Dynasty

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This research project, funded by Taiwan's National Science and Technology Council (NSTC), examines a comprehensive collection of archival documents from the Qing dynasty—the last imperial regime in China—specifically those records relating to local officials' investigations of popular religious activities. The Qing authorities often regarded such popular religious sects as potential threats to state stability, and thus meticulously documented the networks among sect members, geographic distribution of their activities, and their financial resources. Some documents also include verbatim transcripts of interrogations, providing contemporary scholars valuable insights into the sociopolitical and cultural dynamics of the era.

Our objective is to scrutinize the formation and development of these popular religious sects, specifically focusing on whether elements such as kinship, geographical location, or occupational ties played a crucial role in their inception and evolution. The operational mechanisms and mobilization patterns of these sectarian organizations will also undergo meticulous examination. By assessing variations in geographical distribution and tracing developmental trends throughout the Jiaqing era, we aim to explore possible correlations between the rapid proliferation of these secret religious societies and the natural disasters, local disturbances, or regional wars of that period. A central research question is what factors motivated these groups to seek collective redemption or salvation, despite the risk of political repression.

Given that the majority of these archival sources exist only as handwritten manuscripts, a crucial preliminary step involves their conversion into machine-readable formats to facilitate extensive textual analysis. For this purpose, the project employs two platforms developed by the Academia Sinica Center for Digital Cultures: the Optical Character Recognition and Proofreading Platform (Wang 2022), and the Digital Humanities Analysis Platform (Wang / Lee 2021). Additionally, we utilize large language models (LLMs)—ChatGPT-4o and Llama-3.1-70B—to enhance named entity recognition (NER), particularly for accurately identifying personal and place names. This innovative approach significantly enriches the digital humanities methodology, enabling deeper exploration into historical patterns of religious governance and state surveillance practices during the Qing dynasty.

In terms of technical achievements, our optical character recognition (OCR) model can achieve a character error rate (CER) of less than 5% for printed text, but it only achieves a CER of 70% for handwritten cursive text. To address this, we applied fine-tuning techniques and leveraged the context of the text to improve OCR results. These methods successfully reduced the CER for handwritten cursive text from 70% to 20%, which is even lower than that achieved by some research participants. Additionally, we conducted experiments to assess the feasibility of using LLMs to recognize named entities, such as people's names and place names, in text. Our findings showed that by using text chunking and chain-of-thought prompting, both ChatGPT-4 and Llama-3.1-70B achieved F1 scores above 80% for recognizing people's names and over 70% for recognizing place names. The integration of OCR and LLM technology has significantly accelerated our research progress, enabling large-scale text analysis.

Lastly, we reflect on the outcomes of this experimental research thus far, assessing the digital humanities research tools for their limitations and contributions, areas that need further improvement, and potential blind spots within the context of historical research. Concurrently, we also present a forward-looking perspective on the future development of this research project, plan to make the research materials publicly available.

Keywords: digital humanities, large language model, optical character recognition, Qing dynasty, religious societies.

## Bibliography

**Wang, Hsiang-An** (2022): *" Academia Sinica Online Optical Character Recognition and Proofreading System", in: 2022 Pacific Neighborhood Consortium (PNC) Annual Conference and Joint Meetings* , Arizona, USA, September 2022.

**Wang, Hsiang-An. / Lee, You-Sheng.** (2021) **: " The Development and Applications of the Academia Sinica Digital Humanities Research Platform ", in: Journal of Digital Archives and Digital Humanities , 7 : 99 –13 1 .** DOI: 10.6853/DADH.202104_(7).0004